

# Answer key

Form P · Practice

Marking guide and worked solutions · 50 points · may be shared with students after the review

**Read this first.** This practice trains the same skills as the real midterm through different questions and a different kind of result (a surprising sign, two variables that cannot be separated, an extreme day the model cannot explain). Aim to understand why each step works. Stages 1, 6 and parts of the others reward **reasoning defended with a reason or a number**; apply follow-through. Rules of thumb on the reference sheet:  $|z| > 2$  unusual,  $|t| > 2$  significant,  $|r| > 0.7$  heavy overlap. Objective: **Con** concept, **Cal** calculation, **Com** computation in Sheets.

## 1 · Quick answer sheet

| Q  | Objective            | Expected answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Pts | How the marks split                                                                 |
|----|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------------------------------------------------------------------------------------|
| 1  | LO1, LO5 · Con       | gelato_units: <b>DROP</b> (known only at closing; caused by the same weather: a symptom, not a cause). pastry_revenue_eur: <b>DROP</b> (price × units contains the outcome). beach_event: <b>KEEP</b> (crowds nearby buy pastries; known from the calendar). humidity_pct: either, with a reason (weak mechanism for pastries and it overlaps with temperature and rain, or damp days drive indoor comfort food).                                                                     | 4   | 1 per row: decision with a reason from how the café works.                          |
| 2  | LO1 · Con            | Two directional, falsifiable hypotheses on different variables (e.g. “pastry sales are higher on weekends”, “on days with a beach event”, “on cooler days”) and a falsifying LINEST result (opposite sign with $ t  > 2$ , or $ t  < 2$ ).                                                                                                                                                                                                                                            | 3   | 1 + 1 hypotheses (variable, direction, falsifiable) + 1 falsifying result.          |
| 3  | LO2 · Con            | (i) reliability / comparability (the sensor’s conditions changed, so the readings are not comparable); (ii) reliability (repeat counts differ); (iii) validity (pastries sold is not enjoyment).                                                                                                                                                                                                                                                                                      | 2   | 1 for $\geq 2$ of 3 correct + 1 correct explanation of one.                         |
| 4  | LO3, LO4 · Con + Cal | (a) A weekly cycle: the red weekend dots are the high points. Variable: <b>weekend</b> . (b) $z = (103 - 67.07) \div 10.80 = 3.33$ .                                                                                                                                                                                                                                                                                                                                                  | 2   | 1 pattern and variable + 1 z-score.                                                 |
| 5  | LO3, LO5 · Cal + Com | (a) rainfall_mm (mean 0.76 vs median 0.00: strongly right-skewed) and pastry_units (mean 67.07 vs median 65.00: mildly right-skewed; a few high days pull the mean up). (b) Rainfall is a weak predictor ( $r^2 = 0.01$ ); its significance rests on the storm days: without them $ t  < 2$ . It is a fragile predictor of everyday sales. (c) $=IF(OR(B2="Saturday",B2="Sunday"),1,0)$ .                                                                                             | 3   | 1 two variables with the mean/median evidence + 1 reading of the check + 1 formula. |
| 6  | LO3, LO6 · Cal + Con | (a) Residual = $103 - 84 = +19$ , about 2.3 SE: the model cannot explain it. (b) Do not delete without a reason: investigate (a festival tie-in, a promotion, a recording error). Keeping it hardly changes the model (weekend 11.30 vs 10.91); report results with and without. One such day in 168 is not alarming.                                                                                                                                                                 | 2   | 1 residual in SE + 1 defended decision.                                             |
| 7  | LO3 · Com            | (a) weekend, $r = 0.50$ . (b) temperature_c and sunshine_hours ( $r = 0.78$ ); pastry_price_eur and post_change ( $r = 1.00$ ).                                                                                                                                                                                                                                                                                                                                                       | 2   | 1 + 1 (both pairs).                                                                 |
| 8  | LO3, LO5, LO6 · Con  | (a) The pastry price changed only once, on 30 June (no price trials), so price is essentially the date flag. (b) The price effect cannot be separated from anything else that changed on 30 June (season, temperature, events): LINEST cannot tell them apart, so a price coefficient is unstable and imprecise. To estimate it you would need price variation <b>within</b> a period. (c) Keep temperature (stronger $ r $ with pastry_units and a direct mechanism), drop sunshine. | 3   | 1 + 1 + 1.                                                                          |
| 9  | LO3, LO5 · Cal       | $b = 0.50 \times 10.80 \div 0.45 = 11.9$ ; $a = 67.07 - 11.9 \times 0.29 = 63.7$ . Slope: weekend days sell about 12 more pastries than weekdays on average (it equals the difference between the group means, $75.6 - 63.7 = 11.9$ ). Intercept: the weekday average ( $\approx 63.7$ ).                                                                                                                                                                                             | 2   | 1 slope and intercept + 1 meaning.                                                  |
| 10 | LO5, LO6 · Com       | (a) Row 1 reversed: Col 1 = wind_kmh ( $-0.26$ ), Col 2 = beach_event (7.67), Col 3 = weekend (11.30), Col 4 = temperature_c ( $-1.00$ ), Col 5 = intercept (88.43). (b) $t = -1.00 \div 0.19 = -5.3$ : significant; each +1 °C goes with about 1 fewer pastry a day, holding the others constant. Reason: on hot days customers switch to cold items (gelato, iced coffee) or want fewer warm, heavy foods. (c) $df = 163$ in Row 4: $n = df + k + 1 = 163 + 4 + 1 = 168$ .          | 3   | 1 matching + 1 t and interpretation with a reason + 1 n from df.                    |

|    |                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |                                                                                             |
|----|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------------------------------------------------------------------------------------------|
| 11 | LO6 · Cal + Con           | (a) $t = -0.26 \div 0.10 = -2.6$ . (b) Adjusted $R^2$ : $M3 = 1 - (1 - 0.379) \times 167 \div 164 = 0.368$ ; $M4 = 1 - (1 - 0.407) \times 167 \div 163 = 0.392$ . (c) Test it on days the model has not seen: refit with and without wind on the first 112 days and compare the MAE on later days; check the sign is stable in another period; ask whether there is a mechanism (windy days, fewer outdoor customers). Keep it only if it earns its place.                                                                                                                                                                                             | 3 | 1 $t$ + 1 adjusted $R^2$ values + 1 sensible check against chance.                          |
| 12 | LO3, LO6, LO7 · Cal + Con | (a) $SE = \sqrt{(10.9^2/84 + 10.5^2/84)} = 1.66$ ; difference = $-2.9$ ; $t = -1.75$ : not significant. (b) No. "Not significant" means the data cannot distinguish a price effect from chance (absence of evidence is not evidence of absence). (c) Price overlaps almost perfectly with post_change ( $r = 1.00$ ) and moves with the season, so LINEST cannot separate them: the SE (6.48) is larger than the coefficient ( $-5.61$ ).                                                                                                                                                                                                              | 3 | 1 SE and $t$ with decision + 1 + 1.                                                         |
| 13 | LO6 · Cal                 | Jul 16: predicted 63.5, residual $-2.5$ , $ res  < 16.8$ : ordinary. Jun 26: predicted 63.3, residual $-18.3$ , $ res  > 16.8$ : unusual. Investigate before deleting; about 1 day in 20 is beyond 2 SE even if the model is fine.                                                                                                                                                                                                                                                                                                                                                                                                                     | 2 | 1 predictions and residuals + 1 classification and action.                                  |
| 14 | LO4, LO6 · Con            | (a) The biggest miss is Aug 9: actual 103, predicted 82 (under-prediction by about 21). (b) Dots mostly below the line mean the model <b>over-predicts</b> : systematic bias, so something that matters has been left out or has changed since the fitting period.                                                                                                                                                                                                                                                                                                                                                                                     | 2 | 1 + 1.                                                                                      |
| 15 | LO5, LO6 · Cal            | Baseline: Aug 2 67.6, Aug 11 67.0, Aug 19 62.8, Sep 4 68.4.  errors : baseline 12.4, 17.0, 1.8, 9.4 → <b>MAE 10.1</b> ; M3 6.4, 9.9, 2.7, 1.3 → <b>MAE 5.1</b> ; M5 8.0, 5.3, 8.6, 0.1 → <b>MAE 5.5</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                              | 5 | 1 baseline + 2 errors ( $\geq 10$ of 12 = 2, $7-9 = 1$ ) + 2 MAEs (all three = 2, two = 1). |
| 16 | LO6 · Con                 | (a) $R^2$ can only rise when variables are added; it measures fit to days already seen, noise included. (b) M3: its MAE is lower than M5's here (5.1 vs 5.5; 6.9 vs 7.3 over all 56 unseen days) with fewer inputs. Both beat the baseline (8.9 over all 56 days), but the improvement over a weighted-average baseline is modest because the unexplained scatter (SE of $y \approx 8.6$ ) is large relative to the signal.                                                                                                                                                                                                                            | 2 | 1 + 1 (choice, evidence, and the comparison with the baseline).                             |
| 17 | LO7 · Con                 | Plain-language memo: recommends weekend, beach_event and temperature (negative) and why; states a typical error in pastries (SE of $y \approx 9$ , or MAE $\approx 7$ on unseen days); one limitation (one season; weak predictors, $R^2$ only about 0.38; cannot foresee unusual days).                                                                                                                                                                                                                                                                                                                                                               | 3 | 1 + 1 + 1.                                                                                  |
| 18 | LO1, LO6, LO7 · Con       | Two different explanations, e.g. real world: customers switch from pastries to cold items in the heat (substitution); data/other variable: the daily mean temperature may be a poor proxy for the morning hours when pastries sell, or temperature may be standing in for something else that changed over the summer (the 30 June price rise, the mix of customers); adding that variable and watching the temperature coefficient would show it. Plan: add the candidate to LINEST, check $t$ , SE of $y$ and the sign on unseen days; a result that would change the mind (e.g. coefficient near zero, unstable sign or no gain in unseen-day MAE). | 4 | 1 real-world explanation + 1 data/confounding explanation + 1 test plan + 1 falsifier.      |

## 2 · Worked tables

**Q15, hold-out days (Table 6).** Baseline =  $0.5 \times (1 \text{ week ago}) + 0.3 \times (2 \text{ weeks ago}) + 0.2 \times (3 \text{ weeks ago})$ .

| Day    | Actual | Baseline calculation                            | Baseline | M3   | M5   | errors  (base / M3 / M5) |
|--------|--------|-------------------------------------------------|----------|------|------|--------------------------|
| Aug 2  | 80     | $0.5 \times 61 + 0.3 \times 83 + 0.2 \times 61$ | 67.6     | 73.6 | 72.0 | 12.4 / 6.4 / 8.0         |
| Aug 11 | 50     | $0.5 \times 77 + 0.3 \times 59 + 0.2 \times 54$ | 67.0     | 59.9 | 55.3 | 17.0 / 9.9 / 5.3         |
| Aug 19 | 61     | $0.5 \times 71 + 0.3 \times 57 + 0.2 \times 51$ | 62.8     | 58.3 | 52.4 | 1.8 / 2.7 / 8.6          |
| Sep 4  | 59     | $0.5 \times 70 + 0.3 \times 62 + 0.2 \times 74$ | 68.4     | 60.3 | 59.1 | 9.4 / 1.3 / 0.1          |
| MAE    |        |                                                 |          |      |      | <b>10.1 / 5.1 / 5.5</b>  |

## 3 · Where students go wrong

- Reading LINEST Row 1 in the order of the X columns (Q10): it is reversed.
- "Temperature must be positive" (Q10): the data say otherwise; mark the reasoning, not agreement with intuition.
- Reading "not significant" as "no effect" (Q12).
- "M5 is best because  $R^2$  is higher" (Q16).

- **Keeping or deleting the extreme day without looking at the residual (Q6).**